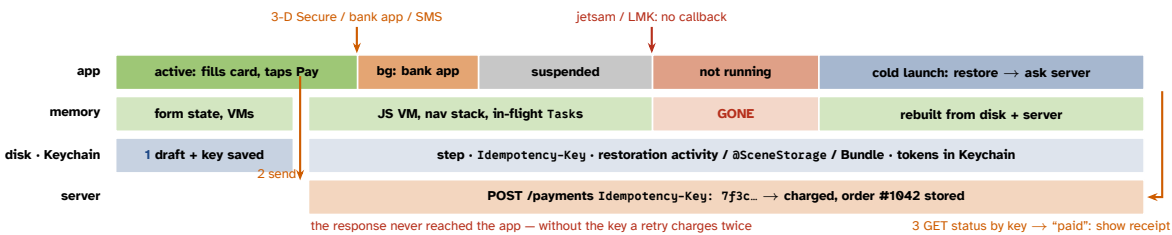


In one line: A backgrounded app is **suspended**, and a suspended app can be **killed without any callback** (iOS jetsam, Android low-memory killer). When the user comes back it is a **cold launch**: everything in **memory is gone**, everything on **disk / Keychain** / **restoration data survives** — and the **server may already have done the payment**. So persist the screen *and* the draft before leaving, and make the operation **idempotent** so resuming can ask “did it happen?” instead of doing it twice.



Ways an app dies

kill	callback?	restore?
iOS jetsam (suspended)	none; applicationWillTerminate only if running in bg	yes: restoration data kept
iOS watchdog / crash	none	crash in restore → data discarded
user swipes app away	none reliable	no — user meant “quit”
Android process death (LMK)	none; onSaveInstanceState ran at onStop	yes: task + Bundles
Android config change	onDestroy + recreate	VM survives (not death)
reboot / OS / app update	none	disk only

- **iOS:** the system relaunches into the saved scene; you rebuild it from stateRestorationActivity / @SceneStorage (small UI state: ids, step, tab) + the draft on disk. Save on sceneDidEnterBackground, not in willTerminate.
- **Android:** the task and back stack come back; the *top* Activity is recreated from its Bundle and a **new** process — Application.onCreate runs again, every singleton and static is fresh. SavedStateHandle / rememberSaveable hold ids; big data goes to disk (Bundle ≈ 1 MB Binder limit).
- **React Native:** the JS VM is gone; the app boots JS from scratch at the initial route. Persist navigation state (React Navigation initialState + onChange) and drafts yourself (AsyncStorage / MMKV; secrets in Keychain/Keystore). On Android, **react-native-screens** discards the restored fragment state (RNScreensFragmentFactory) to avoid crashes — so native restore will not bring your screens back.

What survives

survives	lost
files, DB, UserDefaults already written	objects, caches, singletons, JS state
Keychain / Keystore items	in-flight requests on a default session
restoration data, Bundles (system kill)	timers, Tasks, coroutines, listeners
background URLSession transfers	“we’ll save it in willTerminate”
the server’s side of the request	the response you never received

Test it — on purpose

- **iOS:** background the app, then Stop in Xcode (or xcrun simctl terminate booted com.example.app), relaunch. Simulator *Debug* → *Simulate Memory Warning* tests *warnings*, not death. Swipe-away is *not* the test.
- **Android:** adb shell am kill com.example.app while in the background = real process death; Developer options *Don’t keep activities* only destroys Activities; *Background process limit: none* for soak testing.
- Automate one UI test: fill the form, kill, relaunch, assert the draft.

Design checklist — a death-proof flow

1. Every screen can be rebuilt from **route + ids** — no object handed over in memory that exists nowhere else.
2. Drafts persisted on change (debounced), *before* leaving the app.
3. Side-effecting calls carry an idempotency key; the server has a **status endpoint** for it.
4. Session state re-hydrates lazily (token from Keychain, user from cache) — never “set by the login screen”.
5. Restored ids validated; missing → a sane fallback screen.
6. Drafts cleared on success, cancel and logout (privacy).

Example — a checkout that survives

```
struct CheckoutDraft: Codable {
  var cartID: String, step: Int
  var payKey: UUID? // set once, at "Pay"
}

@MainActor final class Checkout {
  var draft: CheckoutDraft { didSet { store.save(draft) } }
  func pay() async throws { // key saved BEFORE the call
    let key = draft.payKey ?? UUID(); draft.payKey = key
    try await api.pay(draft.cartID, idempotencyKey: key)
  }
  func resume() async throws { // on cold launch
    guard let key = draft.payKey else { return }
    switch try await api.paymentStatus(key) {
    case .paid: route(.receipt); store.clear()
    case .notFound, .failed: route(.pay) // safe to retry
    case .pending: route(.waiting) }
  }
}

<NavigationContainer // React Navigation
  initialState={deepLink ? undefined : saved}
  onChange={s => storage.set("nav", JSON.stringify(s))}>
```

Payments and purchases

- **Idempotency key:** client makes one UUID per *intent*, persists it with the draft, sends it on every retry (Idempotency-Key on POST/PATCH); the server stores the result per key and replays it. On resume: **query status**, never blindly re-POST.
- Server truth wins: order state comes from the server / webhook, not from “the app saw 200”. Redirect flows (3-D Secure, bank app) return via a universal link into a possibly *cold* app — the route must work from zero.
- **StoreKit 2:** a purchase interrupted by a kill is re-delivered through Transaction.updates (listen from launch) until you finish() it — grant content first, then finish.

Traps

- ! Relying on applicationWillTerminate / onDestroy — not called on a system kill of a suspended app.
- ! Restoring into a screen whose data is gone — validate ids, fall back.
- ! Relaunched in the background before first unlock: .complete files and WhenUnlocked Keychain items are unreadable.
- ! Persisting RN nav state that crashes on restore — the user is stuck; reset it from an error boundary.
- ! Android: a singleton filled by the splash/login screen is null when the process is restored straight onto screen 3 — classic NPE.

Remember

Memory dies, disk lives, the server remembers. Save before you send; ask before you retry.